

STEP 1 — Install Required Libraries

```
!pip install gensim numpy pandas matplotlib scikit-learn
import gensim.downloader as api
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.5.1)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.1.2)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
27.9/27.9 MB 46.2 MB/s eta 0:00:00

Installing collected packages: gensim
Successfully installed gensim-4.4.0
```

STEP 2 — Import Libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.manifold import TSNE
```

STEP 3 — Load Pre-trained Embedding Model

```
model = api.load("glove-wiki-gigaword-100")

print("Vocabulary Size:", len(model.index_to_key))

word = "doctor"
print(f"\nVector for '{word}':\n", model[word])
```

Vocabulary Size: 400000

```
Vector for 'doctor':
[ 0.043244 -0.47529  0.15808  0.20413 -0.15383  0.72284  0.26145
 0.20892 -0.3147  -0.070307 -0.43367  0.053109  0.73635  0.98111
 0.23535 -0.10449  0.50258 -0.033356 -0.35537  0.64549 -0.37103
-0.10052 -0.76929 -0.16957 -0.15648  0.53548  0.35146 -1.5126
 0.050984  0.24445 -0.35688  0.43968 -0.62985  0.32891 -0.53009
 0.49832 -1.2061  0.27797  0.42734  0.095773 -0.43527  0.93561
 0.36039 -0.83114  0.12966 -0.1363 -0.58124  0.092946 -0.014708
 0.32562  0.41204  0.1451  0.49803  0.86926 -0.18033 -1.6227
-0.64565  0.17504  0.73849  0.39156  0.83135  0.51308  0.12999
-0.21288  0.68456  0.056297  0.090792  0.28032 -0.12233  0.60761
-0.57913 -0.024127 -0.063252  0.40747  0.10775  0.57977  0.092789
-0.15588 -0.36494 -0.46632  0.37553  0.164 -0.75138 -0.10833
-2.1363 -0.11658  0.23157  0.24425 -0.36352 -0.34928  0.60001
-0.10529  0.78614  0.45707 -0.1712  0.6362  0.68622 -0.28397
-0.29773 -0.81148 ]
```

STEP 4 — Select Word List

```
words = [

    "doctor", "nurse", "hospital", "medicine", "disease",
```

```

"dog", "cat", "lion", "tiger", "elephant",

"computer", "laptop", "keyboard", "internet", "software",

"india", "china", "america", "delhi", "london",

"rice", "pizza", "burger", "apple", "banana",

"happy", "sad", "angry", "joy", "fear",

"car", "bus", "train", "airplane", "bike"
]

```

```

word_vectors = []
valid_words = []

for w in words:
    if w in model:
        word_vectors.append(model[w])
        valid_words.append(w)

word_vectors = np.array(word_vectors)

print("Total selected words:", len(valid_words))

```

Total selected words: 35

STEP 5 — Apply t-SNE

```

tsne = TSNE(n_components=2, random_state=42, perplexity=5)

reduced_vectors = tsne.fit_transform(word_vectors)

print("Shape after reduction:", reduced_vectors.shape)

```

Shape after reduction: (35, 2)

STEP 6 — Plot Visualization

```

plt.figure(figsize=(10, 8))

x = reduced_vectors[:, 0]
y = reduced_vectors[:, 1]

plt.scatter(x, y)

# Annotate words
for i, word in enumerate(valid_words):
    plt.annotate(word, (x[i], y[i]))

plt.title("t-SNE Visualization of Word Embeddings")
plt.xlabel("Dimension 1")
plt.ylabel("Dimension 2")
plt.grid()

plt.show()

```

